COGNIFYZ TECHNOLOGIES

Software Development Internship Guide & Solution Suite

Complete Implementation Guide for Tasks 1 through 6 (Java Full-Stack Portfolio)

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1. About the Internship & Requirements

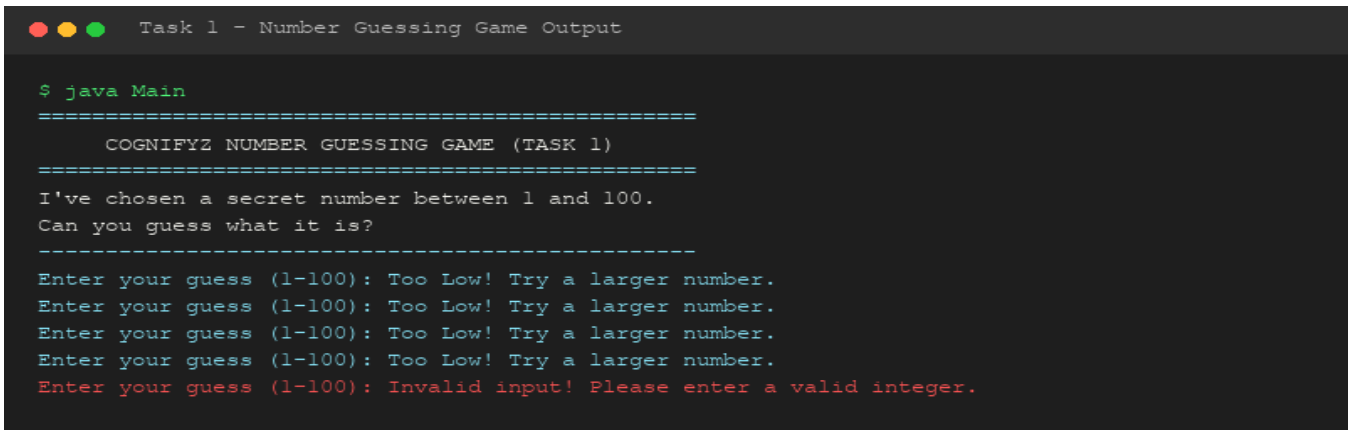
Cognifyz Technologies is an innovative tech company specializing in AI, Machine Learning, Data Analytics, and enterprise technology solutions. The Software Development Internship program consists of 6 structured tasks grouped into 3 distinct difficulty levels. Interns are required to complete at least 4 out of 6 tasks. However, to construct a high-impact GitHub portfolio demonstrating Java proficiency, completing all 6 tasks is strongly recommended.

Level	Task	Difficulty	Project Built	Key Concepts
Level 1	Task 1	Beginner	Text-Based Game	if/else, while, Scanner, Random
Level 1	Task 2	Beginner	Number Patterns	Nested for loops, rows & cols
Level 2	Task 3	Intermediate	Console Task CRUD	OOP, ArrayList, CRUD ops
Level 2	Task 4	Intermediate	Temperature Converter	Arithmetic methods, printf
Level 3	Task 5	Advanced	Persistent Task Manager	File I/O, tasks.txt, Exception handling
Level 3	Task 6	Advanced	Interactive Web Scraper	HttpURLConnection, DOM regex

2. Level 1: Beginner Tasks

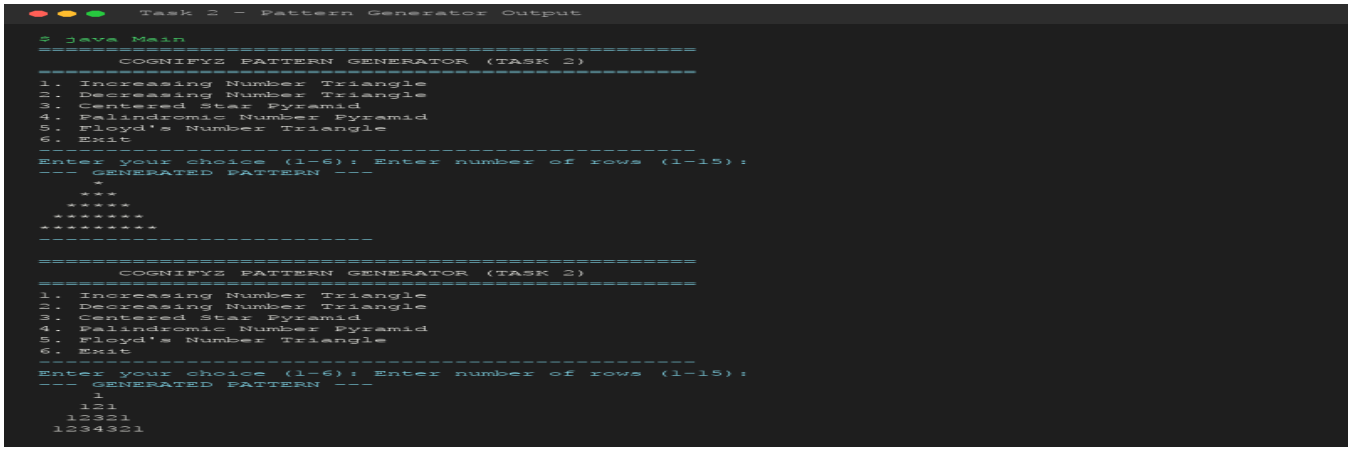
Task 1: Basic Text-Based Game (Number Guessing Game)

Objective: Implement game logic using conditional statements (if, else if, else) and user input loops.
Implementation Details: The computer generates a secret random integer between 1 and 100. The player inputs guesses until correct. The application provides directional feedback ('Too High!' / 'Too Low!'), counts total attempts, and awards a performance rating.



Task 2: Generate and Print Number Patterns

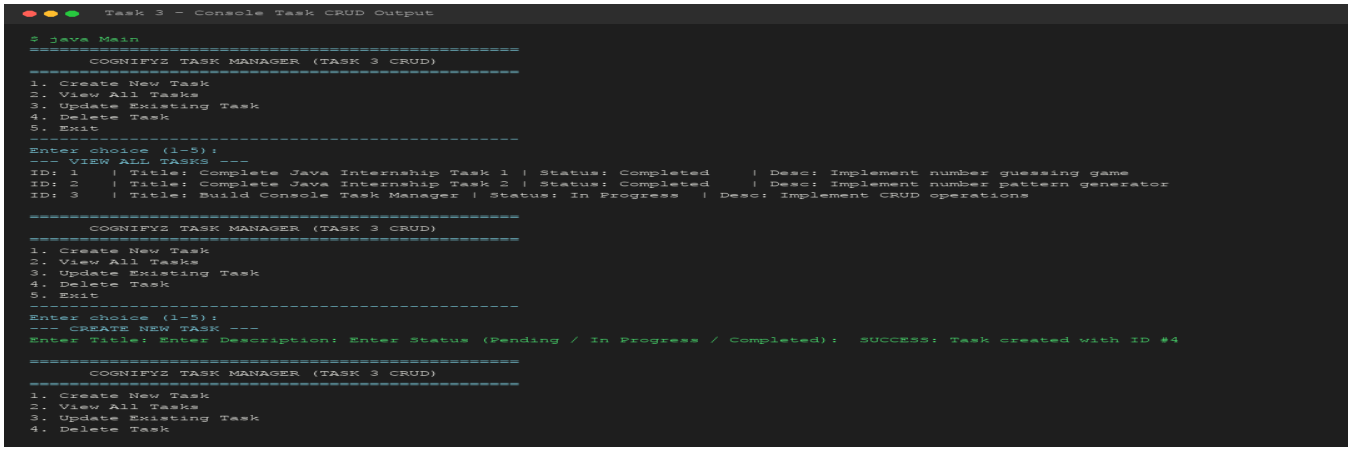
Objective: Utilize nested loops to control the spatial structure of number and star patterns.
Implementation Details: An interactive menu allows users to select from 5 distinct geometric patterns including Increasing Number Triangle, Decreasing Number Triangle, Centered Star Pyramid, Palindromic Pyramid, and Floyd's Triangle.



3. Level 2: Intermediate Tasks

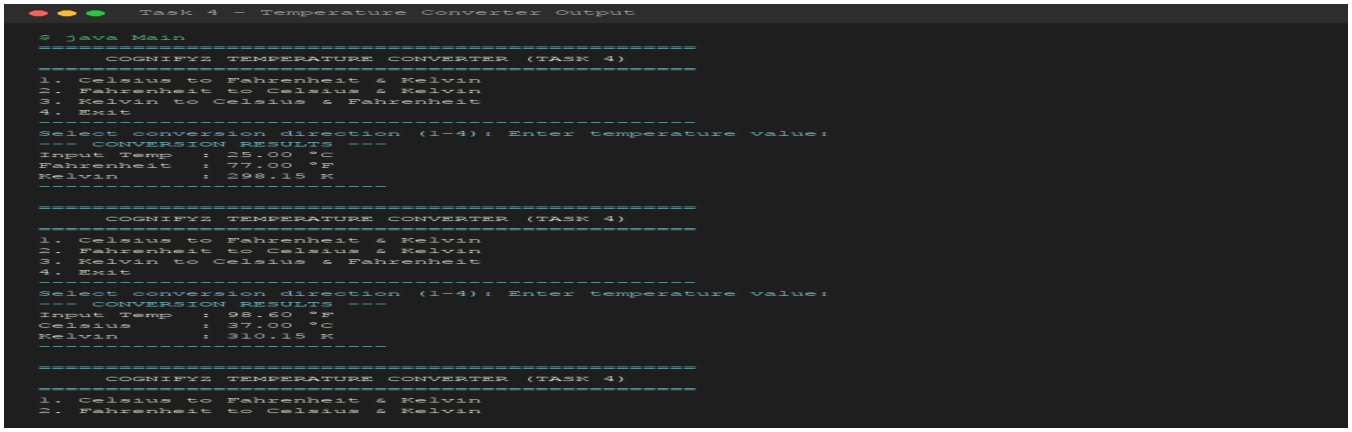
Task 3: Console CRUD Application (Task Manager)

Objective: Implement Create, Read, Update, and Delete operations using object-oriented principles and dynamic lists.
Architecture: Consists of Task.java (encapsulating id, title, description, status), TaskManager.java (encapsulating an ArrayList<Task> data store), and Main.java (CLI menu).



Task 4: Temperature Converter Program

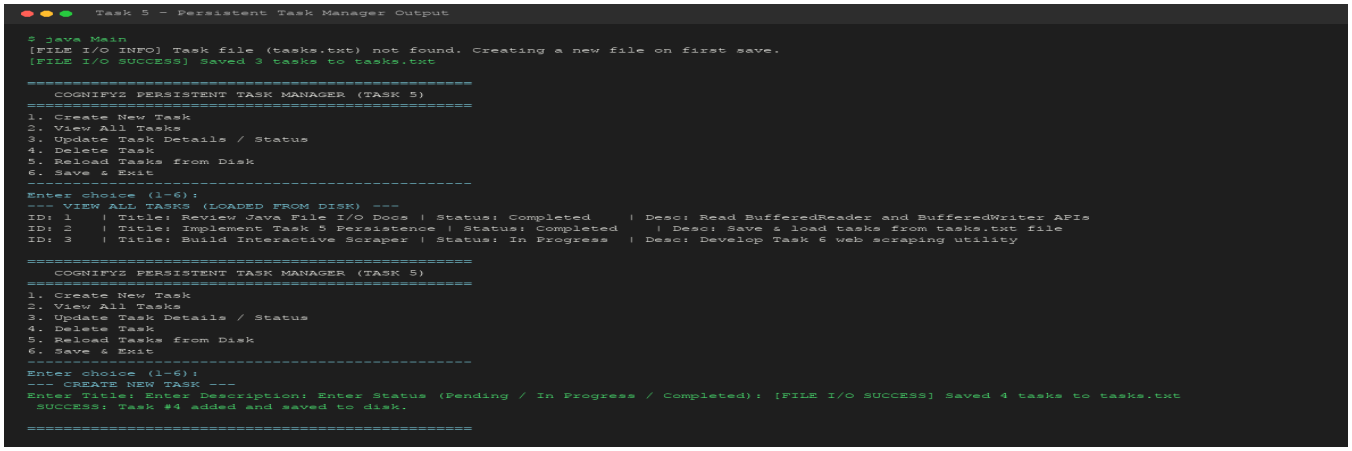
Objective: Convert temperatures between Celsius, Fahrenheit, and Kelvin scales.
Implementation Details: Implements conversion formulas with double precision floating-point arithmetic, user direction selection, formatted tabular output via System.out.printf, and validation for physical limits (Absolute Zero / 0 Kelvin).



4. Level 3: Advanced Tasks

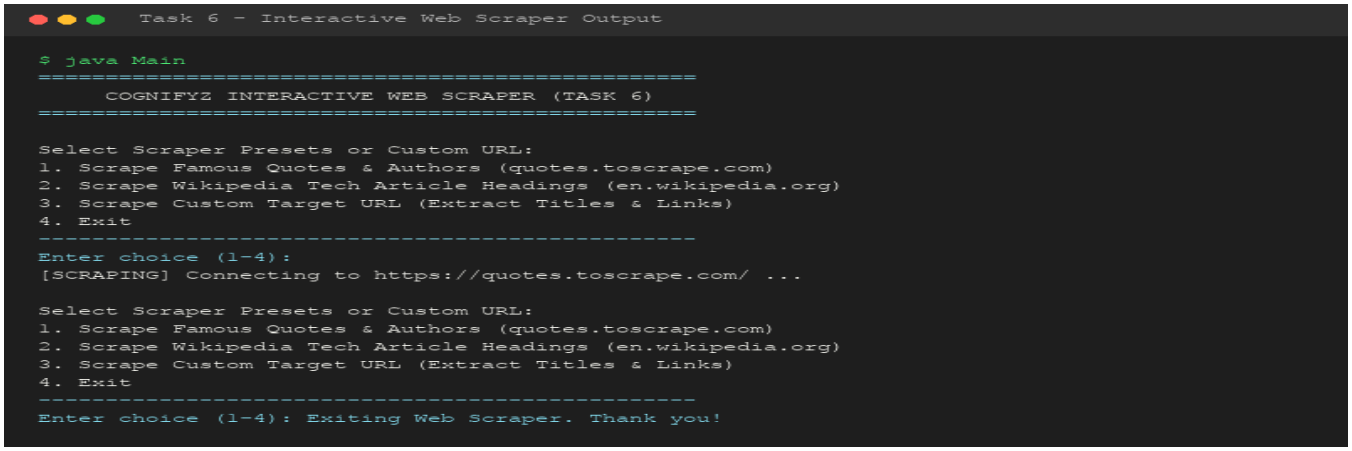
Task 5: Persistent CRUD Application Using File I/O

Objective: Store task data persistently in a local text file (tasks.txt) to maintain state across application restarts.
Implementation Details: Builds directly on Task 3 by introducing TaskFileManager.java. Utilizes BufferedReader and BufferedWriter with pipe-delimited text serialization. Includes robust exception handling for missing files and I/O errors.



Task 6: Interactive Web Scraping Program

Objective: Fetch live web content over HTTP and extract structured HTML elements for user presentation.
Implementation Details: Uses HttpURLConnection to connect to web targets (Quotes to Scrape, Wikipedia, or custom URLs), extracting titles, section headers, quotes, authors, and hyperlinks using Regular Expressions (Pattern / Matcher).



5. Official Submission Requirements & LinkedIn Guidelines

Cognifyz Technologies emphasizes academic integrity and professional presentation. Submitting copied code is strictly prohibited.

LinkedIn Post & Video Guidelines:
• Record a video (~4 mins) demonstrating all completed tasks and showing your GitHub repository.
• Post the video on LinkedIn, tagging @**Cognifyz Technologies**.
• Include official hashtags: **#cognifyz #cognifyzTech #cognifyzTechnologies**.
• Save your project organized cleanly by levels in separate subdirectories.

Final Submission Checklist:

Status	Requirement	Verification
[✓]	Task 1: Text-Based Number Guessing Game	Source, README, Screenshot
[✓]	Task 2: Number & Star Pattern Generator	Source, README, Screenshot
[✓]	Task 3: Console CRUD Task Manager	Source, README, Screenshot
[✓]	Task 4: Temperature Converter Program	Source, README, Screenshot
[✓]	Task 5: Persistent CRUD Task Manager (File I/O)	Source, tasks.txt, README, Screenshot
[✓]	Task 6: Interactive Web Scraper	Source, README, Screenshot
[✓]	Repository README & Documentation	Root README.md & PDF Guide